Exploring Visual In-domain adaptation and In-context Learning

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Abstract

Deep neural networks excel at static classification tasks but often struggle with challenges that biological systems handle efficiently through neuromodulation, such as out-of-distribution (OOD) generalization and online adaptation. In this work, we propose a bio-inspired architectural framework that integrates Adaptive Whitening and Neuromodulatory Gating into standard residual networks. Drawing on the Efficient Coding Hypothesis, our whitening module dynamically decorrelates feature channels during inference to mitigate feature collapse. Furthermore, we implement computational analogues of the Noradrenergic (NA) and Cholinergic (ACh) systems: the former modulates gain based on statistical deviation (signaling "surprise"), while the latter gates features based on input consistency (signaling "reliability"). To bridge these biological priors with modern In-Context Learning (ICL), we analyze attention dynamics in In-Context Robot Transformers (ICRT) and validate our neuromodulated adapters on the Meta-World benchmark. Our results across ImageNet-100, COCO, and robotic tasks demonstrate that these mechanisms significantly improve out-of-distribution robustness and adaptation efficiency. The accompanying code has been made available here.

1 Introduction

Current deep learning lacks the dynamic adaptability of biological systems, limiting autonomous robotics in shifting environments. To bridge this gap, we propose a bio-inspired framework integrating Adaptive Whitening and Neuromodulatory Gating into frozen backbones. Guided by the Efficient Coding Hypothesis, our whitening module dynamically decorrelates features, while Noradrenergic (NA) and Cholinergic (ACh) modules modulate gain based on "surprise" and "reliability" signals. This offers a parameter-efficient solution for Out-of-Distribution (OOD) robustness without expensive retraining.

Parallel to this, we investigate In-Context Learning (ICL) via the In-Context Robot Transformer (ICRT) (Fu et al. [2024]), which performs multi-task imitation by attending to support demonstrations. We analyze ICRT's attention dynamics to understand task selection and concurrently evaluate our neuromodulated adapters on pretrained robotic representations (R3M). By synthesizing these approaches, we demonstrate that biological gain control serves as a powerful, lightweight analogue to attention-based adaptation for robust sensorimotor control.

38th Conference on Neural Information Processing Systems (NeurIPS 2024).

2 Related Works

Out-of-Distribution Detection. Standard deep neural networks often exhibit high confidence on unknown data, a problem formally described as open set recognition Scheirer et al. [2013]. Early baselines utilized the Maximum Softmax Probability (MSP) to detect misclassified and out-of-distribution (OOD) examples Hendrycks and Gimpel [2017]. Subsequent approaches improved upon this by using temperature scaling and input perturbations, such as ODIN Liang et al. [2018]. More recently, energy-based models have emerged as a powerful alternative, demonstrating that mapping the logit distribution to free energy scores provides a superior metric for distinguishing in-distribution (ID) from OOD samples Liu et al. [2020]. Our work builds on these evaluation protocols but introduces architectural modifications rather than just post-hoc scoring rules.

Normalization and Feature Whitening. Deep networks rely heavily on normalization layers for convergence. Batch Normalization Ioffe and Szegedy [2015] centers and scales features but does not decorrelate them. The "Efficient Coding Hypothesis" suggests that biological sensory systems minimize redundancy by whitening inputs Barlow [1961], a principle recently modeled computationally via gain-modulating interneurons Duong et al. [2023]. In this work, we implement adaptive whitening mechanisms that dynamically decorrelate feature channels, extending standard residual networks He et al. [2016] to better handle distribution shifts.

Neuromodulation in Deep Learning. While standard architectures successfully emulate static receptive fields, they often lack the dynamic adaptability of biological brains. Neuromodulation regulates synaptic transmission and plasticity in real-time Hasselmo and Sarter [2011], Mei et al. [2022]. Recent work has begun translating these principles to artificial systems, exploring cell-type specific modulation for credit assignment and using neuromodulatory dynamics to improve structured flexibility in RNNs Costacurta et al. [2024]. Furthermore, multi-neuromodulatory systems have shown promise in enabling adaptive and continuous learning capabilities. We synthesize these insights into computational modules that actively gate features based on "surprise" (NA) and "reliability" (ACh).

3 Model Architecture

The **In-Context Robot Transformer (ICRT)** is a multimodal transformer for multi-task imitation learning (Figure 1), consisting of a vision encoder, modality-specific projection networks for proprioception and actions, and a 12-layer LLaMA-style causal transformer. At each timestep, RGB images from multiple camera views are independently encoded using a shared ViT-Base and aggregated via attention pooling into a fixed-length visual embedding. The robot's 7D proprioceptive state and 7D action vector are projected by separate two-layer MLPs into 768-dimensional embeddings. Two tokens are formed per timestep: a state token obtained by concatenating visual and proprioceptive embeddings, and an action token derived from the previous action. These tokens are appended sequentially to form an alternating state–action sequence, which is processed by the transformer. Outputs at state token positions are decoded by a two-layer MLP ($768 \rightarrow 128 \rightarrow 160$) to predict the next 7D action, comprising end-effector position, orientation, and gripper command. During training, multiple trajectories from the same task are concatenated, with the first k serving as in-context demonstrations, enabling in-context imitation learning across tasks.

4 Method

4.1 Context Window in In-Context Learning

In this experiment, we analyze the attention behavior of the ICRT model when prompted with multiple episodes from either similar or dissimilar tasks using in-context learning.

For each prompt set, we generate and visualize attention maps across all 12 transformer layers of the model. These maps reflect how strongly the query episode attends to each of the support episodes at each layer.

For evaluation purposes, we use the task `real_episode_2024-06-01-tiger-to-black-bowl_34` as the query, and compare its attention against all the episodes passed as context in the prompt.

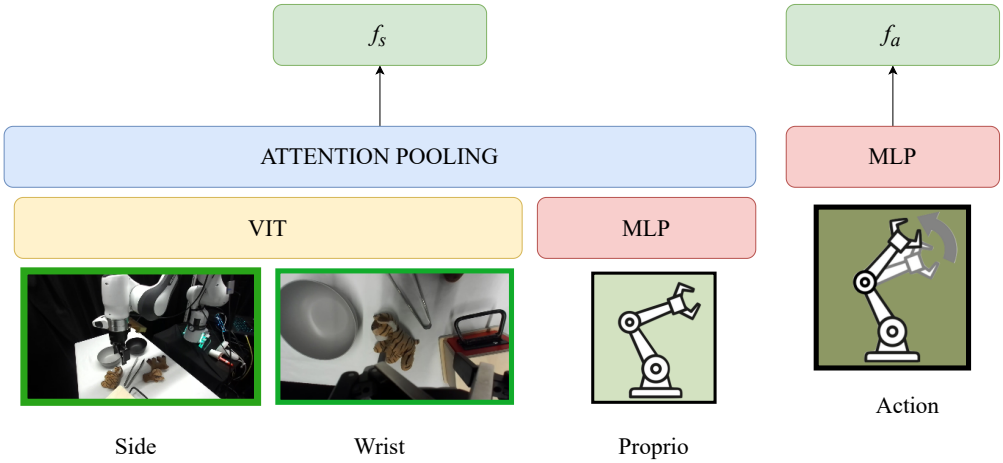


Figure 1: Camera inputs are encoded via a pre-trained Vision Transformer, and proprioception is processed using an MLP. The resulting features are concatenated and passed through attention pooling to extract the state representation f_s. The current action is separately encoded using another MLP to obtain the action feature f_a.

The goal of this analysis is to examine whether meaningful patterns emerge in the attention distribution, such as preferential focus on task-relevant context episodes or consistent attention depth.

4.1.1 Baseline: Prompting Configuration 0

In this baseline experiment, we evaluate the behavior of the ICRT model when provided with only a single support episode as the prompt. Specifically, we use one episode from the task `tiger-to-black-bowl` as input to `icrt.prompt`. We then extract the attention maps from all 12 transformer layers of the model. Since there is only one support episode in the prompt, the attention maps reflect how the query episode attends to itself across the layers.

4.1.2 Prompting Configuration - 1

Tasks Used:

Task 1: close-drawer
Task 2: open-drawer

Here we provide the ICRT model with 20 episodes drawn from two different tasks (Task 1 to Task 4) each as the prompt input. These support episodes are passed via `icrt.prompt`, following the in-context imitation learning framework described in the paper *In-Context Imitation Learning via Next-Token Prediction*.

We then extract attention maps from each of the 12 transformer layers of the model. Each attention map shows how one episode attends to all other episodes in the combined prompt set.

After obtaining the episode-level attention maps, we apply spectral clustering to analyze the internal structure of the attention patterns. This allows us to identify latent groupings among the in-context episodes based on how strongly they attend to one another. By examining both the resulting cluster structure and off-diagonal attention patterns, we can assess whether the model selectively relies on a subset of episodes while down-weighting others. In particular, this analysis helps distinguish episodes that consistently receive high attention and appear to play a critical role in guiding the model's predictions from episodes that contribute marginal or redundant information. Such episodes, while present in the prompt, may add extraneous context without materially influencing the model's behavior. Overall, spectral clustering provides a principled way to probe how the transformer internally organizes and prioritizes in-context demonstrations beyond what is visible from raw attention maps alone.

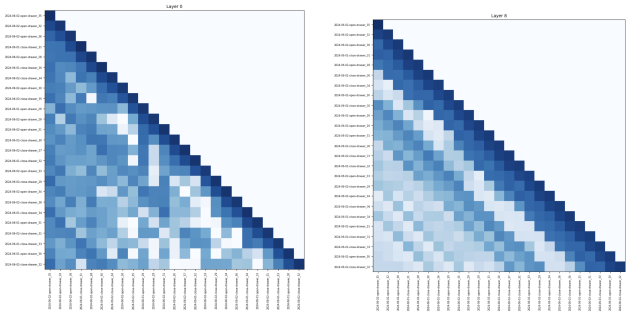


Figure 2: Episode-level attention maps at early (Layer 0) and late (Layer 8) transformer layers.

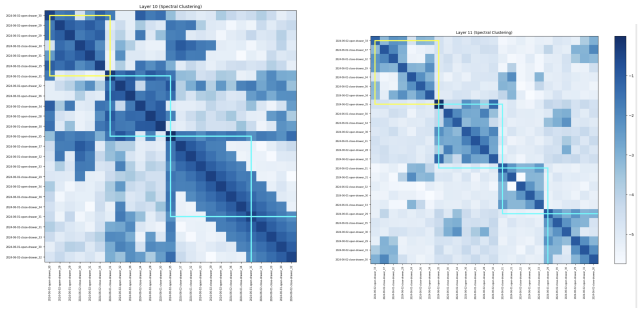
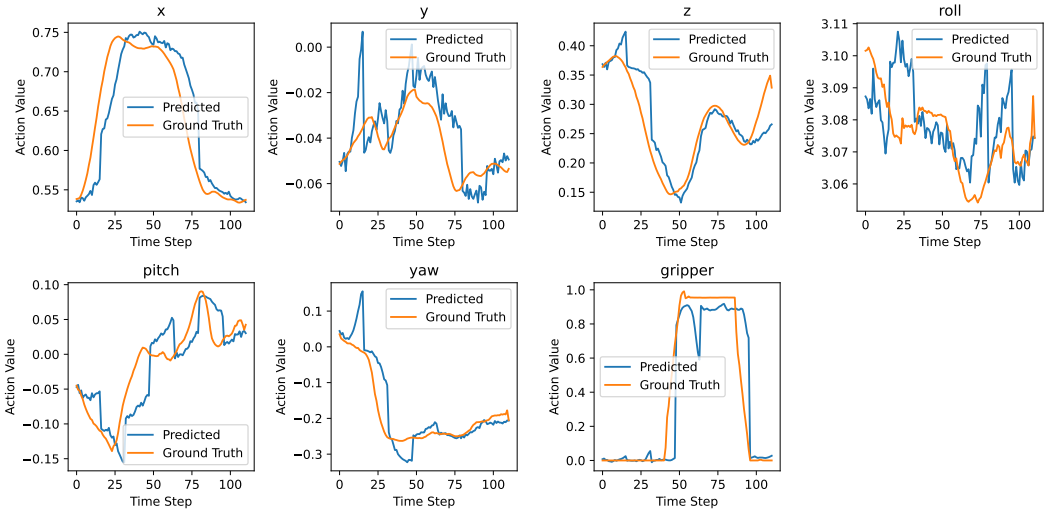


Figure 3: Episode-level attention maps after spectral clustering

Results for Prompting Configuration 1

After providing the model with episodes from *Task 1* and *Task 2*, we select a subset of episodes identified as important based on the spectral clustering analysis of the attention maps. Using only these selected episodes as in-context demonstrations, we then evaluate the model on the `Close Drawer` task in a previously unseen environment. Figure **??** shows the comparison between the predicted actions and the ground-truth actions.



The results indicate that the model implicitly infers which episodes from Task 1 and Task 2 are most relevant for solving the target task, and preferentially attends to these episodes during prediction. This observation suggests that, rather than selecting in-context episodes at random, it is beneficial to first analyze how episodes influence one another through attention and then construct the context

using only the most informative demonstrations. Such a selection strategy leads to more effective use of the context window and improved generalization to unseen environments.

4.2 Neuromodulation-based adaptation

We adopt a frozen visual backbone (ResNet-50 or ViT) as the perceptual front-end of an in-context learning (ICL) agent, where the backbone provides a stable representation and downstream behavior is adapted on-the-fly from context (few demonstrations, recent frames, or short interaction history) rather than by updating backbone weights. To test the hypothesis that neuromodulatory-style gain control can improve robustness under distribution shift and make frozen representations more context-adaptable, we implement three biologically-inspired, lightweight modules on top of the backbone: adaptive whitening (to reduce spurious low-level correlations), noradrenaline-style signal enhancement (to amplify informative deviations), and acetylcholine-style reliability gating (to suppress unreliable features). Across all settings, backbone weights remain frozen; only the linear/MLP readout and (when enabled) the modulation parameters are learned, making these modules a parameter-efficient analogue of context-conditioned adaptation in robot perception.

4.2.1 Adaptive Whitening

Motivated by adaptive whitening mechanisms in early visual cortex (Duong et al. [2023]), this module aims to remove domain- and sensor-specific correlations that can interfere with ICL. In a humanoid setting, such correlations arise from changing illumination, motion blur, exposure, camera intrinsics, or background statistics; decorrelating early features can improve both novelty handling and the stability of context-based adaptation.

Given input $\mathbf{x} \in \mathbb{R}^{B\times C\times H\times W}$, we first flatten it to $\tilde{\mathbf{x}} \in \mathbb{R}^{N\times C}$ where $N = B \cdot H \cdot W$, then compute:

$$\mathbf{M} = \left(\mathbf{I} + \mathbf{W}\,\mathrm{diag}(\mathrm{softplus}(\mathbf{g}))\,\mathbf{W}^\top\right)^{-1} \tag{1}$$

where $\mathbf{W} \in \mathbb{R}^{C\times C}$ is a fixed random orthogonal projection and $\mathbf{g} \in \mathbb{R}^C$ are adaptive gains. The whitened output is $\tilde{\mathbf{y}} = \tilde{\mathbf{x}}\mathbf{M}^\top$, reshaped back to the original dimensions. The final output blends original and whitened features:

$$\mathbf{y} = (1-\sigma)\cdot\mathbf{x} + \sigma\cdot\tilde{\mathbf{y}} \tag{2}$$

where $\sigma \in (0,1)$ is the blending strength. At test/deployment time, gains $\mathbf{g}$ adapt via $\mathbf{g} \leftarrow \mathbf{g} + \eta(\bar{\sigma}^2 - 1)$, where $\bar{\sigma}^2$ tracks running projection variances. We view this as a minimal, online "context" signal that continuously calibrates early features to the current stream without updating backbone weights.

4.2.2 Noradrenaline (NA) Modulation

Inspired by NA's role in sensory cortex (Mei et al. [2022]), this module emphasizes informative deviations in feature statistics while de-emphasizing random fluctuations. In the ICL framing, NA modulation acts like a context-sensitive "novelty/saliency" controller: it amplifies channels whose current statistics depart from the expected regime in a structured way, helping the downstream in-context readout separate task-relevant changes (new object, new viewpoint, new contact state) from nuisance variation.

For each channel c, we compute spatial statistics $\mu_c = \mathrm{mean}_{H,W}(x_c)$ and $\sigma_c^2 = \mathrm{var}_{H,W}(x_c)$, then measure deviations from running estimates of "typical" in-stream behavior:

$$\Delta_\mu^{(c)} = \frac{\mu_c - \bar{\mu}_c}{\sqrt{\bar{\sigma}_c^2 + \epsilon}}, \quad \Delta_\sigma^{(c)} = \frac{\sigma_c^2 - \bar{\sigma}_c^2}{\bar{\sigma}_c^2 + \epsilon} \tag{3}$$

where $\bar{\mu}_c, \bar{\sigma}_c^2$ are exponential moving averages updated online at test/deployment time. A learned saliency network $f_\theta : \mathbb{R}^{2C} \to [0,1]^C$ predicts which channels carry informative deviations, producing multiplicative gains:

$$\mathbf{g}_{\mathrm{NA}} = \mathrm{softplus}\left(\mathbf{b} + f_\theta([\Delta_\mu, \Delta_\sigma]) \odot |\Delta_\mu|\right) \tag{4}$$

where $\mathbf{b} \in \mathbb{R}^C$ is a learnable per-channel baseline gain. The modulated output is $\mathbf{y} = \mathbf{x} \odot \mathbf{g}_{\mathrm{NA}}$.

4.2.3 Acetylcholine (ACh) Modulation

Motivated by cholinergic modulation of attention and learning (Mei et al. [2022]), this module suppresses channels deemed unreliable under the current context. For ICL in humanoid robots, reliability can vary with occlusion, glare, motion blur, depth/texture ambiguity, or viewpoint-induced aliasing; gating unreliable channels can stabilize both OOD rejection and context-based adaptation by discouraging the readout from overfitting to spurious evidence in a small context set.

A network $r_\phi : \mathbb{R}^C \to [0,1]^C$ estimates per-channel reliability from global-average-pooled features:

$$\mathbf{g}_{\mathrm{ACh}} = \mathbb{1}[r_\phi(\mathrm{GAP}(\mathbf{x})) > \tau] \cdot r_\phi(\mathrm{GAP}(\mathbf{x})) \tag{5}$$

where τ is a learnable threshold. Soft gating (sigmoid relaxation) is used during training; soft gating and hard gating ($\mathbb{1}[\cdot]$) are applied at test time. In the ICL view, ACh implements a lightweight attentional "mask" that biases the frozen representation toward channels that remain consistent across a short context window.

4.2.4 Integration with frozen backbones (ResNet-50 and ViT)

Early whitening with parallel modulation. Based on preliminary findings that whitening benefits concentrate in early processing, whitening is restricted to early stages (ResNet stem and `layer1`; for ViT, the patch embedding / earliest transformer block), while NA/ACh modulation is applied in later stages. In all cases, the backbone remains frozen, and these modules provide a controllable interface for context-dependent adaptation without relearning the representation.

ViT integration. For ViT backbones, we apply the same conceptual operations to token representations: whitening is applied to early token/channel statistics to remove low-level correlation structure, while NA/ACh apply per-channel gains to token features (computed from token-level or pooled statistics) to implement saliency amplification and reliability gating. This keeps the experimental comparison aligned across architectures: a frozen foundation encoder with a small set of trainable modulation parameters that can be interpreted as a mechanistic analogue of in-context adaptation.

4.3 R3M-based visuomotor policy with neuromodulation

Apart from analysis of in-context learning models, we also test our neuromodulation-based adapters on the R3M (Nair et al. [2023]) model which was pretrained on the Ego4D (Grauman et al. [2022]) dataset. We use a pretrained R3M ResNet-50 visual encoder, resize to 244, and wrap it with optional neuromodulation blocks. A behavior-cloning (BC) policy maps $(\text{image}, \text{state}) \mapsto \text{action}$ by concatenating a 2048-D image embedding with a learned 768-D state projection (state dim 7) and predicting a 4-D action with an MLP head.

5 Experiments

5.1 Gain Modulation for Near- and Far-OOD Recognition as a Proxy for In-Context Learning

We study neuromodulatory-style gain control as a proxy for in-context learning (ICL) in humanoid robot perception. Our hypothesis is twofold: (i) lightweight gain modulation (NA/ACh), when applied to an otherwise frozen visual backbone, can significantly affect robustness under distribution shift, and (ii) restricting whitening to early layers primarily improves robustness by removing low-level, dataset- or sensor-specific correlations. In the robotics setting, this mirrors a common deployment scenario where a humanoid robot relies on a strong frozen visual encoder and must adapt online to new objects or environments using limited contextual information, such as recent observations or demonstrations. Gain modules provide a parameter-efficient mechanism for such context-conditioned adaptation, while few-shot and OOD evaluations serve as operational proxies for in-context learning ability.

We use an ImageNet-pretrained ResNet-50 with Batch Normalization (He et al. [2016], Ioffe and Szegedy [2015]), freezing all backbone weights and training only a linear classifier on global-average-pooled features using SGD (learning rate 10^{-3}). When enabled, gain-modulation parameters are co-trained. We evaluate the baseline, whitening-only, NA-only, ACh-only, and their compositions (NA→ACh, ACh→NA, and NA||ACh). Whitening is applied only to the stem and first stage, while

later stages use gain modulation alone, reflecting a realistic humanoid perception stack where only shallow adaptation is feasible.

Experiments are conducted under two normalization regimes: *BN-frozen*, where the backbone remains in evaluation mode with fixed statistics, and *BN-updating*, where BatchNorm statistics update without gradient flow. BN-updating consistently yields stronger near-OOD performance and is used for far-OOD experiments. In a robotics context, BN-updating approximates online perceptual calibration to the robot's current sensor stream, while BN-frozen corresponds to fully static deployment.

Training is performed on ImageNet100[1] (Deng et al. [2009]), using 70 classes as known and 30 held out as unknown, forming a near-OOD open-set recognition setting (Scheirer et al. [2013]). For far-OOD evaluation, models trained on the same 70 classes are tested on COCO2017 validation images (Lin et al. [2014]), treating all samples as OOD. This results in three setups: (1) ImageNet100 near-OOD with BN frozen, (2) ImageNet100 near-OOD with BN updating, and (3) COCO far-OOD with BN updating. We repeat setups (2) and (3) using a frozen DINOv3 ViT-B/16 backbone, training only the readout and enabled modulation parameters while allowing token-level normalization statistics to update.

Let $z(x) \in \mathbb{R}^{70}$ denote logits over known classes. We compute the Maximum Softmax Probability (MSP) (Hendrycks and Gimpel [2017]) and Energy scores (Liu et al. [2020]) for OOD detection. Performance is evaluated using Top-1 accuracy on known classes, and AUROC, AUPR (known as positive), and $\mathrm{FPR@95\%\,TPR}$ (Liang et al. [2018]) for OOD detection. In the humanoid setting, low $\mathrm{FPR@95}$ is critical to avoid confident misclassification of unfamiliar objects.

To further characterize representations from an ICL perspective, we report mean intra-class and inter-class distances, their ratio, covariance deviation $\|\Sigma - I\|_F$, and k-shot linear probing accuracy on unknown classes for $k \in \{1, 3, 5, 10, 15\}$. These metrics assess how readily a fixed representation can be re-purposed using limited context, serving as a minimal proxy for in-context adaptation.

5.2 Short-MetaWorld-VLA Behavior Cloning with In-Context Evaluation

We train behavior cloning policies on the Short-MetaWorld-VLA dataset using a deterministic $90/10$ train/validation split and R3M-style preprocessing. Two models are trained for 100 epochs (batch size 128): a baseline without neuromodulation and a model with Whitening+NA+ACh enabled in parallel. Both use BN-updating and mixed precision. Performance is measured using validation MSE on predicted actions. To explicitly probe in-context learning behavior, we additionally evaluate an in-context ridge regression readout by fitting a linear map from context features to actions and measuring query MSE on held-out trajectories.

6 Results and Discussion

6.1 Near- and far-OOD recognition results, interpreted for in-context learning in humanoid robots

We report results for the three setups in Experiments. Abbreviations: baseline, W (whitening), NA (Noradrenaline), ACh (Acetylcholine). For clarity of expression, we have only included the most important and insightful results from our experiments.

Near-OOD open-set recognition (BN frozen). Whitening largely preserved closed-set accuracy (baseline 95.5% vs. W 95.1%) while staying close on detection, whereas neuromodulation generally harmed open-set rejection, especially sequential variants. Few-shot transfer was weaker at low k but converged by 15 shots. For humanoid robots, this BN-frozen regime resembles deploying a perception model without any online calibration; in that case, naive gain compositions can degrade open-set rejection, implying that context-like modulation must be designed carefully to avoid overconfident misrecognition of novel objects.

Near-OOD open-set recognition (BN updating). With lightweight on-stream calibration, in-domain accuracy remained similar, whitening was best/tied-best on detection, and gain modulation improved low-shot probing (notably at 1-shot) with convergence by 15 shots. In the humanoid ICL

[1] https://www.kaggle.com/datasets/ambityga/imagenet100

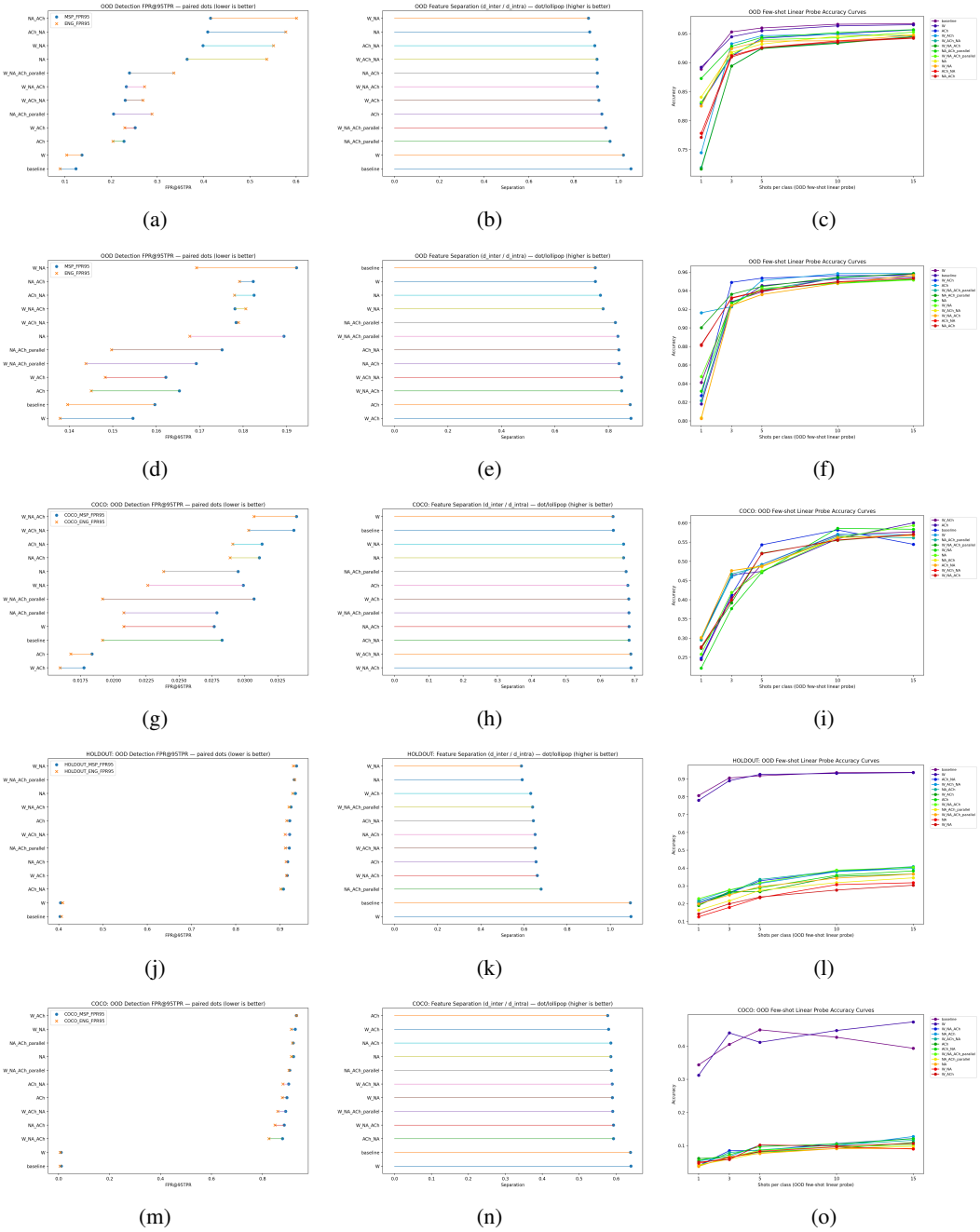


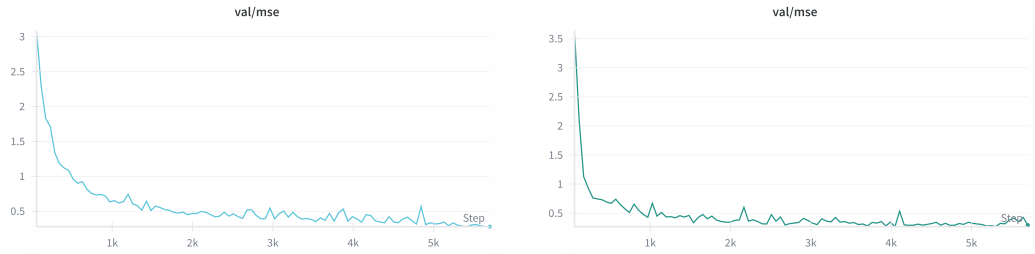
Figure 4: Near- and far-OOD performance across three ResNet setups and two ViT extensions. Rows 1–3 (top to bottom): Setup 1 (BN frozen) with a frozen ResNet backbone and trained linear head evaluated on ImageNet100 near-OOD; Setup 2 (BN updating) with BN running statistics updated during training and evaluated on ImageNet100 near-OOD; Setup 3 (BN updating) with in-domain training on ImageNet100 (70 classes) and far-OOD testing on COCO2017 val. Row 4: ViT analogue of Setup 2 (BN updating) evaluated on ImageNet100 holdout (near-OOD). Row 5: ViT analogue of Setup 3 (BN updating) evaluated on COCO2017 val (far-OOD). Columns (left to right): OOD detection $\mathrm{FPR@95\%\,TPR}$ (MSP and energy), feature separation $d_{\text{inter}}/d_{\text{intra}}$, and few-shot linear probe accuracy vs. shots.

interpretation, BN-updating acts like lightweight on-stream calibration, and the improved low-shot probing suggests that appropriate gain modulation can make representations more context-adaptable, i.e., the robot can incorporate a small number of context examples/demonstrations to form a better decision boundary for new categories.

Far-OOD detection on COCO (BN updating). OOD rejection was already strong, but neuromodulation (especially ACh and W+ACh) improved few-shot repurposing over baseline, including at 1-shot and through 15 shots. For humanoid robots operating in broad, uncurated environments, this suggests a practical takeaway: even when OOD rejection is already strong, gain modulation can reshape geometry and improve few-shot repurposing, which are properties aligned with in-context competence (rapidly learning "what to attend to" from minimal contextual cues).

ViT (BN updating): holdout and far-OOD. Baseline and whitening preserved in-domain accuracy and consistently delivered the best OOD rejection and few-shot probing, while gain-modulated variants collapsed both detection and adaptation (low-shot and at 15 shots). In this regime, whitening is the safer route for improving in-context adaptation without destabilizing OOD behavior.

6.2 Neuromodulated-R3M evaluation results



(a) (b)

Figure 5: Validation mean-squared error (MSE) over training epochs for behavior cloning with an R3M visual backbone. Left: baseline (no neuromodulation). Right: whitening + NA + ACh applied in parallel (W+NA+ACh (parallel)). Lower is better.

As shown in Figure 5, the validation mean-squared error drops faster for the neuromodulated version as compared to the baseline version, demonstrating the quick learning capability imparted by neuromodulation-based adapters. Additionally, upon evaluation using in-context ridge regression, baseline model obtained an MSE value of 3.11, while the neuromodulated-R3M achieved an MSE value of 1.02. This clearly shows that our proposed neuromodulation method is an highly effective and efficient method to make pretrained models more adaptable to changing stimuli. This has direct applications in robotics, where quick response generation and adaptation to the visual environment is very important.

7 Conclusion

We presented a parameter-efficient framework that leverages biological principles of efficient coding and neuromodulation to enhance the adaptability of frozen visual backbones. Our extensive evaluation reveals a distinct functional split: while adaptive whitening stabilizes low-level feature geometry to improve out-of-distribution detection , neuromodulatory gating (NA and ACh) is crucial for enabling rapid few-shot adaptation and reshaping the latent space for in-context tasks.

Furthermore, the integration of these modules into the R3M backbone for robotic manipulation yielded faster convergence in behavior cloning tasks compared to standard baselines. These findings suggest that mimicking biological gain control provides a scalable pathway for creating robust, context-aware robotic perception systems capable of adapting on-the-fly to changing visual stimuli and task demands.

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A Delineation of roles

Vijay Jayawant Harkare. Involved in the ideation, experimental design, and implementation of neuromodulation-based experiments. Designed the novel neuromodulation module, and ran hundreds of experiments to test validity using standardized models and metrics.

Rivujit Das. I study the effect of context window size on in-context learning using the official ICRT codebase from In-Context Imitation Learning via Next-Token Prediction. I apply spectral clustering to attention patterns to analyze how in-context learning evolves as multiple task episodes are added to the prompt, and to identify which episodes contribute most effectively to the model's predictions.

Saurabh Velankar. Involved in the integration of the novel neuromodulation module with standard pretrained models like VIP, R3M, and BCIB, and also involved in the subsequent experimentation. Due to constraints on resources and time, only one model experimentation was included in this study.